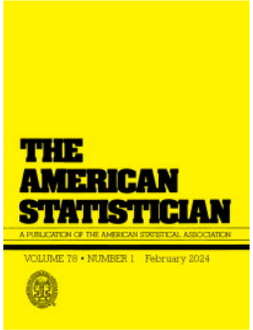

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Play Call Strategies and Modeling for Target Outcomes in Football

Preston Biro and Stephen G. Walker

Department of Statistics and Data Sciences, University of Texas at Austin, Austin, TX

ABSTRACT
This article considers one-off actions for a football coach who is asking for a specific outcome from a play. This will be in the form of a minimum gain in yards, usually in order to gain a first down. Using a random utility model approach we propose the play to be called is the one which maximizes the probability of the desired outcome. We specifically focus on pass plays, which requires the modeling of outcomes in terms of yards gained, for which we use the family of generalized gamma distributions. The data and results relate to the Fall 2021 Presbyterian College football team, in which we leverage specific information pertaining to the offensive playbook.

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1. Introduction

Data-driven decision making in sports is a fairly recent area of active research. Due to its rapid progress, some intuition and suitability of the analytics created have often been criticized due to the models not matching the structure of the game or the inability of analysts to communicate the models effectively. One example of the problem is the football play recommendation literature. There are by now a number of methods developed with the intent of informing a coach when to choose a certain action; for example, whether it be to kick or attempt to convert on a fourth down (Romer 2002; Burke 2016; Lopez 2020; Allen 2021), to attempt a two point conversion (Porter 1967; Sackrowitz 2012), whether to run or pass (Stern 1998; Reed, Critchfield, and Martens 2006; Bradley 2018), or even full play recommendation systems (Patek and Bertsekas 1998; Biro and Walker 2022).

To be more explicit, the criticism usually boils down to one of two points. The first is an issue of complexity: the model does not account for variables which coaches feel are important. For example, many papers addressing the fourth down situation will almost always make a recommendation to go for significantly more attempts where the distance to go is 1 yard or less. However, authors rarely provide results that do not primarily rely on a simple set of game state information, such as distance to go, line of scrimmage, time remaining, and score differential. In turn, they rarely address factors critical to coaches, such as the positioning of the defensive line, injuries sustained by the offense over the course of the game, or even some of the more attainable attributes such as whether or not the team had attempted a fourth down earlier in the game, and in turn the result of any previous attempts.

The second issue is that of mindset: the models prescribe actions that do not align with the coach's goals. For example, many models provide recommendations that aim to maximize some measurement such as expected points or win probability. While recommendations that lead to the maximization of these quantities are likely optimal as long-term behavior, they ignore the play-to-play structure of the game which calls for a more detailed approach. For example, in Week 10 of the 2021 NFL regular season in a game between the Washington Football Team and the Tampa Bay Buccaneers, Washington was faced with a third down and 5 situation on Tampa Bay's 15 yard line. With just over 3 min remaining and only a 4 point lead, Washington faced a critical situation where they had to decide how to maintain or extend their lead, whereas a mistake could lead to a loss. A play calling recommendation system built to optimize expected points or win probability might have recommended a direct attempt to the end zone in order to try to extend the team's lead to two scores (as a touchdown would have increased their lead to at least 10). Alternatively, a recommendation system may have called for a run play, which would reduce the time remaining in the game and set up a risky but manageable fourth down situation, giving their own team an extra attempt to keep possession of the ball and their opponent less time to respond upon failure. However, Washington opted to throw the ball to their best receiver on a 5 yard slant, completing the pass for just enough yards to earn a first down. We can infer from the result that the play call was likely chosen by the coach to maximize the team's chances of earning a first down on that specific play, rather than attempting to maximize their scoring probability or set up a reasonable 4th down. The coach's mindset adapted to the game situation, and thus the policy for evaluation and recommendation should also adapt. Having a single policy overlooks these nuances, and therefore failing to acknowledge these nuances will validate the criticism.

In this article, we aim to address this second critique. The first critique, while valid, can often be addressed with more data, more computation power, and more detailed analyses ensuring that necessary information is included in the model. Here, we

CONTACT Preston Biro prestonbiro@gmail.com; Stephen G. Walker s.g.walker@math.utexas.edu Department of Statistics and Data Sciences, University of Texas at Austin, Austin, TX.
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will address the notion that for a specific play the coach requires a specified outcome, such as gaining a minimum number of yards to retain possession. We require a model which mirrors the mindset of coaches, providing short-term utility adaptations that can be made to address individual concerns or desires on a play-to-play level. Specifically, we will formulate and appropriately adapt a Random Utility Model (RUM) (McFadden 1973; Horowitz et al. 1994) that can be tailored to address the utility structure at any point in a game.

The article will focus on one-off decision making using RUMs. Here, we leverage the idea that under a mindset aiming to achieve a desired outcome, the utility structure can be simplified to a binary choice (success or failure) and therefore the optimal policy entails maximizing the probability of achieving the desired outcome. While the application of this work will be within the context of American Football, one does not need extensive knowledge of the game to understand the approach, and the methods described can be applied in a multitude of alternative settings.

The article arose as a consequence of one of the authors (PB) having first hand experience of working with a football coach, Kevin Kelley, who served as the Head Coach for the Presbyterian Football Team during the Fall 2021 season. Coach Kelley graciously allowed PB to conduct research during the football season, gathering data and learning to understand the mindset of the coach. While the results were not explicitly implemented and tested during the football season, many were discussed with the coach. Presbyterian College's (PC) offense finished third ranked out of over 100 FCS schools in 2021 (The Athletic Staff 2021) despite having the smallest annual enrollment (Haley 2021) of all FCS schools, demonstrating that our guides have mindsets worth emulating. This work will depend heavily on using information pertaining to the team's specific offensive playbook, which until now has been an unexplored research topic in the football play recommendation literature, to our knowledge.

1.1. Related Literature

The optimization of decision making in the game of football has been attempted in many forms. One of the most notable and perhaps earliest of such was that of Porter (1967). The article detailed extra-point strategy after a touchdown is scored. For each examined case, the author uses decision theory to determine the expected utility and corresponding best action. He eventually comes to the conclusion that the optimal strategies in some cases are rarely implemented at the NFL level.

Larger scale play recommendations followed in Stern's chapter of "Statistics in Sports" (Stern 1998), where he laid out a dynamic programming approach to calculating win probability. For a subset of plays near the end of the game, Stern used backward induction to recursively calculate win probabilities, identifying the optimal action to be taken at each subsequent step, choosing between options such as run, short pass, long pass, field goal, and punt. Romer (2002) took another dynamic programming approach, opting to study expected points rather than win probability. Under his model, he was able to analyze fourth down decision making using the expected points that would result from different actions, particularly looking at kicking versus attempting to pick up a first down. Other similar work includes Krasker's dynamic programming win probability (Krasker 2004), Burke's Fourth Down Bot and Expected Points Added (Burke 2016, 2010), Yurko, Ventura, and Horowitz's expected points, win probability, and nflWAR modeling (Yurko, Ventura, and Horowitz 2019), Goldner's decision theory approach to future expected scores (Goldner 2017), Yam and Lopez's causal inference approach to fourth down decision making (Yam and Lopez 2019), Lopez's tracking data examination of the fourth down decision literature (Lopez 2020), and Patek and Bertsekas' approximate policy iteration model for action optimization (Patek and Bertsekas 1998).

The present article can be seen as a development of previous work appearing in Biro and Walker (2022), in which reinforcement learning algorithms identify long-term optimal policies for play calls. The combination of the estimation of probabilities of outcomes for each type of play and the specification of utility functions are the key requirements. The work required improved modeling methods, which was later developed for run plays in Biro and Walker (2023). Similar modeling methods were developed for passing plays in Aubet and Ehrlich (2022) and Ehrlich, Callot, and Aubet (2021), but require the use of data unavailable at the start of the play and therefore are not suitable for our purposes.

While we maintain the notion of maximization of expected utility, this is more a consequence of our formulation which is grounded in Random Utility Models (RUM), and the binary nature of the outcome, that is, success/failure. RUMs are most popularly employed in economics, identifying underlying factors in observed decision making, along with the value associated with changes in certain factors. One example is that of McFadden (1986), who presents methodology using the RUM framework which allows for forecasting observed choice behavior through the specification of the utility function. This allows for improved analysis of market factors, providing insight on the functions which control decision making, such as features of the product or demographics. Horowitz et al. (1994) detail many of the updates to RUM. The main advances are those in computational methods for model evaluation, namely multinomial probit models and Bayesian estimation. The intractability of models once created issues under numerical techniques, but sampling based algorithms provided tractability. Additionally, the task of semiparametric estimation has been accurately accomplished in, for example, Powell, Stock, and Stoker (1989); Fox (2007). Others have attempted to quantify success in the game of football using alternative methods, the most popular being Carroll, Palmer, and Thorn's metric (Carroll, Palmer, and Thorn 1988) which determines success as a function of the down and the distance to first down. However, we will take an alternative binary approach, where the threshold for success is determined by the coach.

Statistically driven decision making has taken place in other sports. In hockey, Beaudoin and Swartz (2010) advocate for more aggressive decisions in willingness to pull the goalkeeper, using data to suggest that the losing team will score more goals and generate more penalties when following such strategy, and doing so earlier in the game will increase simulated win probability. In basketball, Gibbs, Elmore, and Fosdick (2021)

explore the causal effect of calling a timeout to stop a scoring streak of an opponent, determining that the standard practice of using timeouts for these purposes actually has a slightly negative average treatment effect, implying different strategies would be more appropriate. In baseball, Bouzarth et al. (2021) advocate for a statistically driven defensive positioning of players to limit opposing batters from reaching base (known as employing a "shift"), a strategy so successful that it will be banned in the MLB starting in the 2023 season (Miller 2022).

Finally, the role of the expert in analysis is one worth understanding. Brownstein et al. (2019) explore this topic in detail, coming to the conclusion that expert judgment should play a critical role in statistical inference and evidence-based decision-making. This comes from both subject-matter experts and from statistical experts. Subject-matter experts provide critical knowledge that can inform all stages of the scientific process, and can often go a step further by helping to produce probability distributions for representation of uncertain quantities. Statisticians provide an understanding of the effects of modeling and study design elements on the final results, and thus the authors suggest statisticians should play a more critical role in the collaboration with experts in projects involving data. In this project, we will show how the collaboration of a subject-matter expert (a football coach) with statisticians can create a useful tool for both parties involved.

The rest of this article will detail as follows. Section 2 will discuss the statistical motivations for the problem and present the RUM framework. Section 3 will detail necessary modeling efforts. In Section 4 we will present and analyze our results. We will finish with discussion and conclusions in Section 5.

2. Outline of Methodology

In this section we will provide a description of the methods used for a coach to set a particular call which is required to achieve a minimum outcome in terms of yards gained.

2.1. The Action Space

The data comes from the 11 PC football games from the Fall 2021 season. The plays are specific to this offense, designed by Coach Kelley. Each play is represented by situation, the specific formation and play call, and the result of the play in yards gained. We additionally collected several other pieces of relevant information (penalties, pass results, pre-snap motion, etc.). There are 1101 plays in this dataset and Table 1 shows a subset of the information collected and used. Note that the names of plays have been adjusted to limit exposure of the private playbook.

For this task, we will restrict our data to designed pass plays, specifically those that cannot be classified as screens, RPOs (run/pass options), or pseudo run plays, such as a pass thrown behind the line of scrimmage. While these plays are important, their properties are more in line with run plays (and in some cases are classified as run plays in terms of officiating and record keeping) and thus need separate modeling techniques to properly represent. We will also remove plays nullified by penalty, leaving a total of 568 pass plays. Penalized plays can be included in the future, but require some nuances in terms of the transition structure due to the effects of automatic first downs or repeating previous downs. We do include intended pass plays that result in a quarterback scramble, as scrambles are relevant features of pass plays that should be modeled under this structure. Finally, we only use covariates that are known with certainty when the play call is chosen, in order to provide recommendations that would be relevant for decision makers. While information such as defensive coverage would be helpful in predictive modeling, that information would not be available for the play caller and therefore will be not used here.

Table 1. Table of example plays.

DOWN	DIST	LOS	FORMATION	PLAYCALL	GAIN	CMT
3	6	72	2x2 T	SCORPION	11	C
2	20	85	2x3	VOLCANO	0	I
1	10	30	2x3	FREEZER	17	C
1	10	75	2x1	BRANCH	6	C
3	10	12	1x4	SUNSHINE	12	C, TD
...	...	...	...	...	...	...

NOTE: DOWN is the down, DIST is the distance to 1st down, LOS is the line of scrimmage measured from 1 to 99 in distance to scoring an offensive touchdown, FORMATION is the offensive alignment, PLAYCALL is the play called by the coach, GAIN is the amount of yards gained, CMT is a label indicating other relevant play result information, such as a completion (C), incompletion (I), and touchdown (TD), among others. There are other rows and columns in the data not displayed in this table.

2.2. Random Utility Model

In contrast to the expected utility model—a description of how individuals *should* make a rational decision—the RUM is a description of how individuals may *actually* make decisions in practice. To illustrate, suppose there are two possible actions, a and b, and (U_a, U_b) are the utility of outcomes for each. An individual is assumed to choose action a with probability $p_a = P(U_a > U_b)$. The main idea is that there is uncertainty as to which is the best action. This uncertainty is represented by the joint distribution of (U_a, U_b), which can be estimated from available data. The individual believes the best action is a with probability p_a and so the individual rationally chooses action a with probability p_a.

We will take the modeling of outcomes using data from the RUM approach; for each action $(a_j)_{j=1:M}$ we construct the joint model for $(Y_1, \ldots, Y_M)$: the outcomes in terms of random yards gained from each action. This joint density will be of the form $p(y_1, \ldots, y_M \mid \widehat{\theta})$ where the unknown is the finite dimensional parameter θ which is estimated from the data and labeled as $\widehat{\theta}$.

In this formulation we will define the one-off utility in terms of the coach's required outcome, specifically the number of yards the offense needs to gain for the coach to consider the play to be a success, which we will refer to as the threshold gain x. The appropriate utility here will be binary, set to 1 when the outcome of the play in yards gained reaches the prescribed threshold x, and is 0 otherwise, that is,

$$U_x(Y) = \begin{cases} 1 & Y \geq x \\ 0 & Y < x. \end{cases} \tag{1}$$

The choice of a binary (specifically 0 and 1 vs. other values) utility will prove to provide helpful mathematical properties, but is ultimately arbitrary and could be scaled and shifted without

loss of generality. We now have the joint probability model for $(Y_j)_{j=1:M}$ and hence for each a we have the probabilities

$$p_a = P\Big(U_x(Y_a) \geq U_x(Y_b) \quad \text{for all} \quad b\Big).$$

While the choice of taking action a with probability p_a makes sense, it is difficult to persuade individuals of the merit of making random choices, even if this is in reality how individuals do behave. Here we take an alternative decision theoretical perspective. If a hypothetical group of individuals with the above model were making random decisions based on the allocated probabilities, the largest "vote" would be for the action which maximized the (p_a). Hence, assuming the coach would act democratically, he/she would select the action which maximizes

$$V_x(a) = \mathrm{P}(Y_a \geq x). \tag{2}$$

Thus, the value of a particular action is the probability that the desired threshold is realized. This is an appealing notion, which is that the coach will select the action which maximizes the probability of achieving what is required. Under the RUM, we are assuming that the coach is choosing an action that he/she believes maximizes their chance of achieving the stated goal, and he/she is picking an action that would be chosen democratically by a hypothetical group of experts behaving according to the (p_a). We now need to model the outcomes Y_a for each action a.

3. Pass Play Modeling

In this section we detail how we model the pass plays. Each pass play will indexed by i. Line of scrimmage and the gain in yards on play i will be denoted by L_i and Y_i, respectively, and for each i we aim to model the distribution of Y_i. It is important to have accurate estimates for the distribution of outcomes for every play called. A coach's playbook is often quite large and the number of plays actually called in a game is relatively small. In the past, these issues have usually been dealt with by ignoring the specifics of individual plays, opting to treat all pass plays as equal or to classify plays as "short" or "long" passes; see Stern (1998). This has obvious issues, specifically that there are many nuances to each play, and bulk grouping even as short and long can ignore the fact that most plays are not designed to be exclusively one or the other. We will develop a model that allows for the estimation of a probability distribution that properly represents any pass play, and even for those that have not actually been observed in a game.

3.1. Overview of Pass Plays

On any particular pass play, one of five things could occur:

- the ball is thrown and completed,
- the ball is thrown and falls incomplete,
- the ball is thrown and is intercepted,
- the quarterback is tackled for a sack, or
- the quarterback scrambles.

Some of these options have clear outcomes in terms of the yards gained. In particular, an incomplete pass is always a zero yard gain, and an interception is always recorded as such, although the yard line for defensive recovery is variable. For our purposes, we will treat all negative plays (incompletions, interceptions, and sacks) as the same, assuming the desire of the coach is not met on these plays no matter the specific outcome. Conversely, the other two outcomes do not have clear outcomes in terms of yards gained, generally having a distribution of outcomes that varies by play.

Further, the event of a completed pass can be decomposed. We expect different route structures to have different distributions, as balls thrown downfield are more likely to gain a greater number of yards. However, a ball thrown downfield is also more likely to be off-target, and is therefore more likely to be incomplete or intercepted. Thus, we should incorporate route structure when considering the probability of each event occurring.

Finally, just as we do not expect each route to have the same distribution of yards gained, we do not expect each route to be targeted the same on every play. A post route on one play might be the main target, but on another play it might just be a backside route meant to hold a safety. Thus, the probability of each route being targeted should be modeled for each play. The modeling structure required to accurately represent the data generation process is clearly quite complicated, but an accurate representation of each pass play is necessary to properly analyze a scheme.

3.2. The Route Level

We will start by modeling the result of completed pass plays. At its core, a pass play is a combination of several routes. A route is defined as the intended path an eligible receiver runs on a pass play in order to potentially receive the ball. Several different events can happen when a receiver is targeted on a route, the most notable being the completed pass. When completed, the number of yards gained on the route follows a continuous and smooth distribution, and thus we require a density function to represent such results. For this, we choose the generalized gamma family of distributions, so as to model means, variances, and skewness adequately. While the values of yards gained are recorded discretely, we find that since the underlying data generating process is continuous, a continuous density is more appropriate for modeling. Furthermore, this decision would allow for easy extensions to tracking data with increased precision.

The original formulation of the generalized gamma distribution is from Stacy (1962). The generalization is formed by providing a positive exponent parameter to the exponential parameter of the gamma distribution, obtaining the following density function:

$$f(y|a,d,p) = \frac{p/a^d}{\Gamma(d/p)} y^{d-1} e^{-(y/a)^p}, \quad y > 0$$

where $\Gamma(\cdot)$ denotes the usual gamma function.

The generalized gamma distribution has three parameters, (a, d, p); it is smooth in nature, bounded at zero for $d > 1$, unimodal, and has a single heavy tail on its positive end. The relative weight of the tail can be jointly controlled by all parameters, but as Figure 1 indicates, it is done primarily through a and p. The

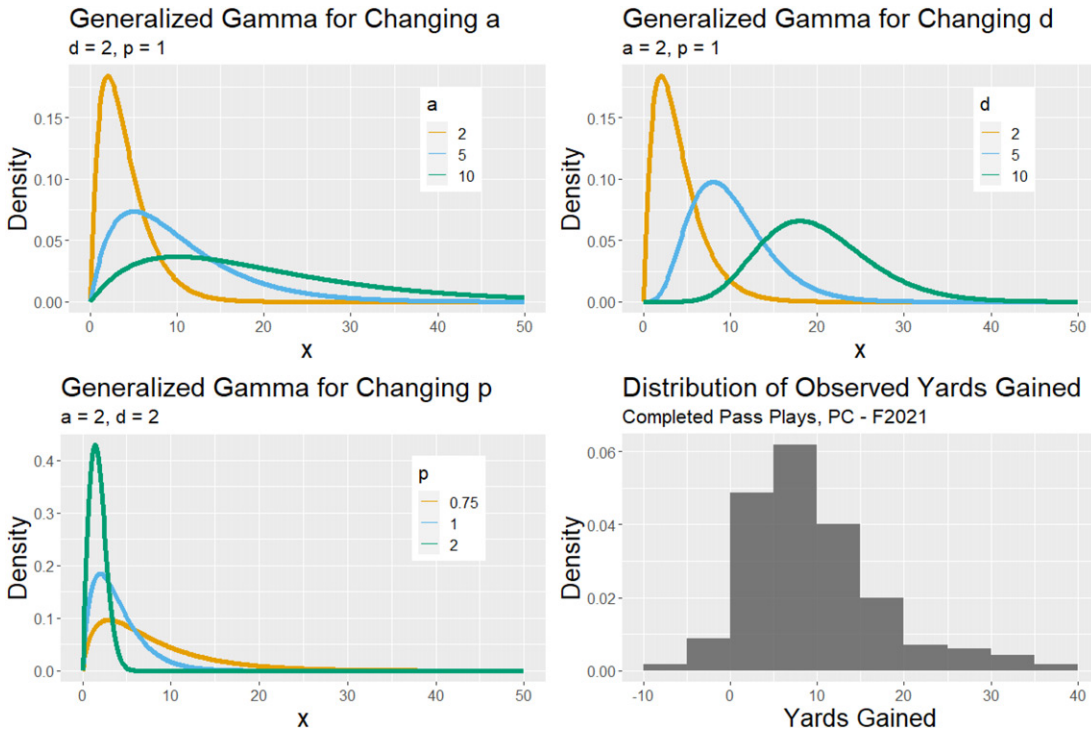

Figure 1. Plots of generalized gamma densities for changing parameter values, along with the distribution of yards gained on all completed (non-touchdown) passes.

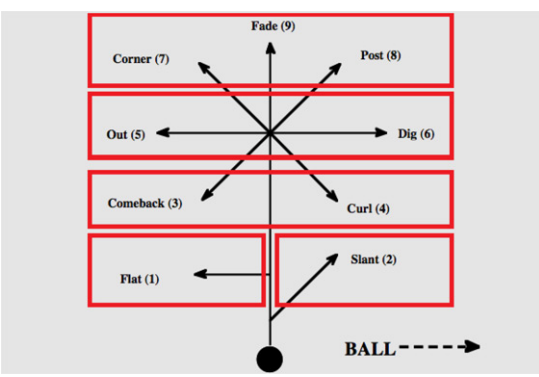

Figure 2. An example wide receiver route tree. An individual route can be seen in this tree by tracing from the black circle to the end of an arrow, which would correspond to receiver's movement from the start of a play to their ending location. For example, an Out route would correspond to a receiver starting at the black circle (the line of scrimmage), moving forward toward the opponent's end zone, and then breaking perpendicularly in the direction opposite of where the ball was at the start of the play. Here, the boxes represent routes that are grouped together. The routes shown do not encompass all possible routes represented in the data.

shapes of the generalized gamma distributions appear to match those of the completed pass data, also displayed in Figure 1.

Moving toward modeling the distribution for each route, a reasonable question is why there is a need to model routes separately from one another. The reason is that different routes require receivers to run at different lengths, depths, directions, speeds, and angles (see Figure 2 for an example route tree, Bowen (2014): standard set of routes for a wide receiver). The different characteristics of each route should determine the yards gained on the route. For example, we would expect a route with greater depth to have a greater amount of yards gained on average than one with a shallower depth, given the pass is completed. We might also expect that an in- or out-breaking route would have a larger spread of outcomes than a stationary route, as a receiver moving in stride at the point of the catch may be more likely to gain additional yards. Thus, by treating each pass play as a mixture of different routes, we can capture more complex behavior in the resulting distribution, such as multimodality that could naturally arise from the structure of the routes involved.

However, there are too few completions observed (279) to model each of the 60 unique routes individually. Thus, we will need to group routes based on common attributes which we believe to have similar enough traits to share a common distribution of outcomes. We choose to put each route into one of five categories:

- SHORT_IN: Short (7 yards or less), in-breaking routes (ex. Slant)
- SHORT_OUT: Short, out-breaking routes (ex. Flat)
- SHORT_STATIONARY: Short, stationary routes (ex. Curl)
- MED: Medium (15 yards or less), non-vertically breaking routes (ex. Dig)
- DEEP_VERTICAL: Deep (more than 15 yards) or vertical routes (ex. Post).

Once grouped, the task of modeling the distributions becomes easier, fitting each group with a generalized gamma density (GG):

$$f(y|\theta_j) \sim GG(a_j, d_j, p_j), \quad a_j \sim C^+(0,2),$$
$$d_j \sim C^+(0,1), \quad p_j \sim C^+(0,1). \tag{3}$$

The priors for each parameter for each of the five settings are based on the positive half-Cauchy distribution, sometimes referred to as a horseshoe prior (Feller 1971; Carvalho, Polson, and Scott 2009, 2010). Additionally, $(\theta_j)_{j=1:5}$ is the vector of parameters (a_j, d_j, p_j) for the corresponding $j \in E$, where E is

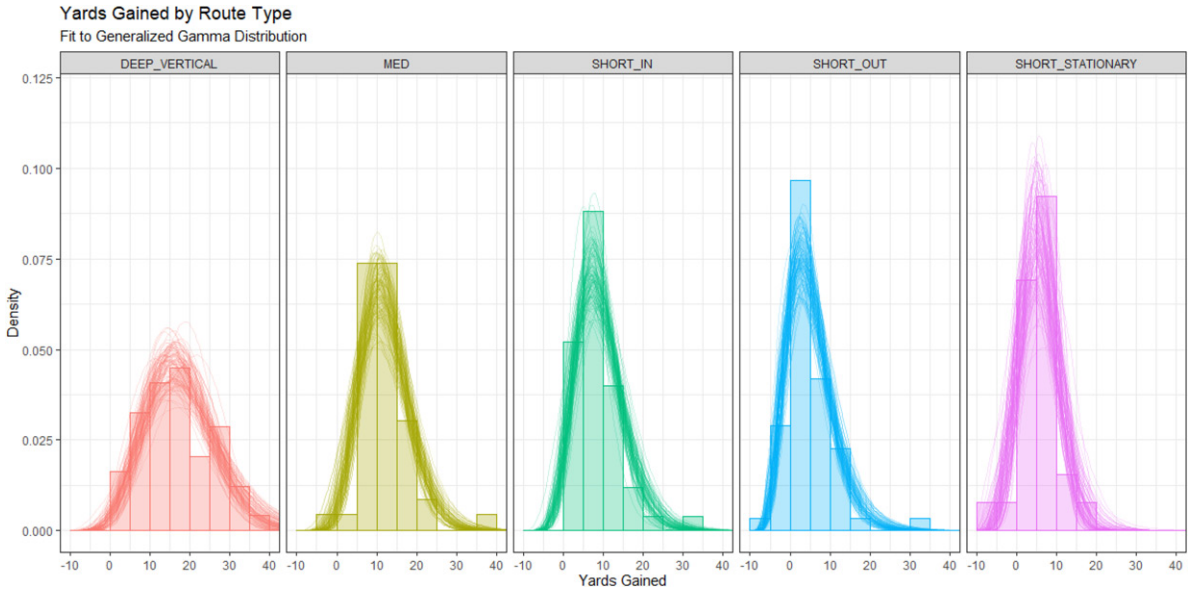

Figure 3. Posterior distributions of route fits for each route grouping.

the set of the five route archetypes. Some of these routes had completed pass observations with gains less than the typical minimum for a generalized gamma distribution of zero, and thus the data were shifted to fit the parameters. Additionally, after observing the structure of yards gained on scramble plays, we noticed they possessed similar structure, and thus they too were fit using a generalized gamma density function.

Figure 3 shows several draws from the fitted posterior predictive generalized gamma distributions for each route grouping, along with the relevant completed passes for each route. The fitting process was completed using a Metropolis-Hastings Gibbs sampler (Hastings 1970; Geman and Geman 1984; Smith and Roberts 1993; Tierney 1994). For each set of parameters, 10000 draws from the posteriors were saved (after burn-in and thinning out) to ease future computations.

3.3. Event Prediction

Predicting the result of a pass play requires much more than understanding the result of completed passes. Pass plays naturally break up into separate events which have distinct outcomes. We will break event prediction into three different levels:

- given the play is an intended pass, whether the ball is thrown, the quarterback is sacked, or the quarterback scrambles;
- given the ball is thrown, which route is targeted; and
- given a specific route (type) is targeted, whether the pass is completed, falls incomplete, is intercepted, or completed for a touchdown.

For each grouping, we assume a Dirichlet-Multinomial conjugate distribution, allowing the mixing parameters to have Dirichlet posteriors with parameters governed by the data and a prior. Priors must be selected carefully to ensure the sparsity of the data does not over-influence the predictive distributions. Therefore, we use priors for the weights based on the number of observations for each category over all plays, a mild form of Empirical Bayes prior (Robbins 1992), shrinking the trends of the mixing parameters to behave closer to the average of observations.

Here we describe the event model for the data. For each play call $i \in P$, with $|P| = 281$, we obtain the posterior probability for outcomes; $w_i = (w_{0i}, w_{si}, w_{ti}, w_{ri})$, where w_0 is for no gain (or negative outcomes) in yards from outcomes incomplete, interception, and sack, w_s is for outcome scramble, w_t for touchdown, and w_r for routes. These latter probabilities are comprised of a conditional probability vector; $w_r = (w_{r_j})_{j=1}^{5}$ where the $j = 1, \ldots, 5$ indexes the five route types. So there are 8 weights all combined. Over a season the raw count of such outcomes over all plays is available and we take the common prior for all weights to be Dirichlet with parameter values coming from the number of outcomes of each type. Formally, the posterior weights are distributed as

$$w_i \sim \mathrm{Dir}(n_{i1} + cN_1, \ldots, n_{i8} + cN_8) \quad \text{for all} \quad i \in P. \tag{4}$$

Here, n_{ij} is the number of times play i had event outcome j (no/negative gain, scramble, touchdown, or route), N_j is the total number of times outcome j occurred over all pass plays, and c is a scaling factor, chosen to be small (in our case, $c = 0.05$) to allow for individual play trends to control their own resulting distribution, and in an attempt to not be overly informative in the prior. Results are robust to this choice for a similar range of c values.

For the hyperparameter for the route target mixture, we use knowledge of the play design to estimate the a priori target share. For this, we discussed with members of the Presbyterian College football team (Coach Kelley, offensive assistant coaches, and quarterbacks) to determine who we should expect to receive the ball on each play, and with what relative proportion. We translated these descriptions into numbers, taking into account the design of the play, the chance for the play to break down, and potential play modifications. The resulting values chosen for the a priori target share range from 1% to 96%.

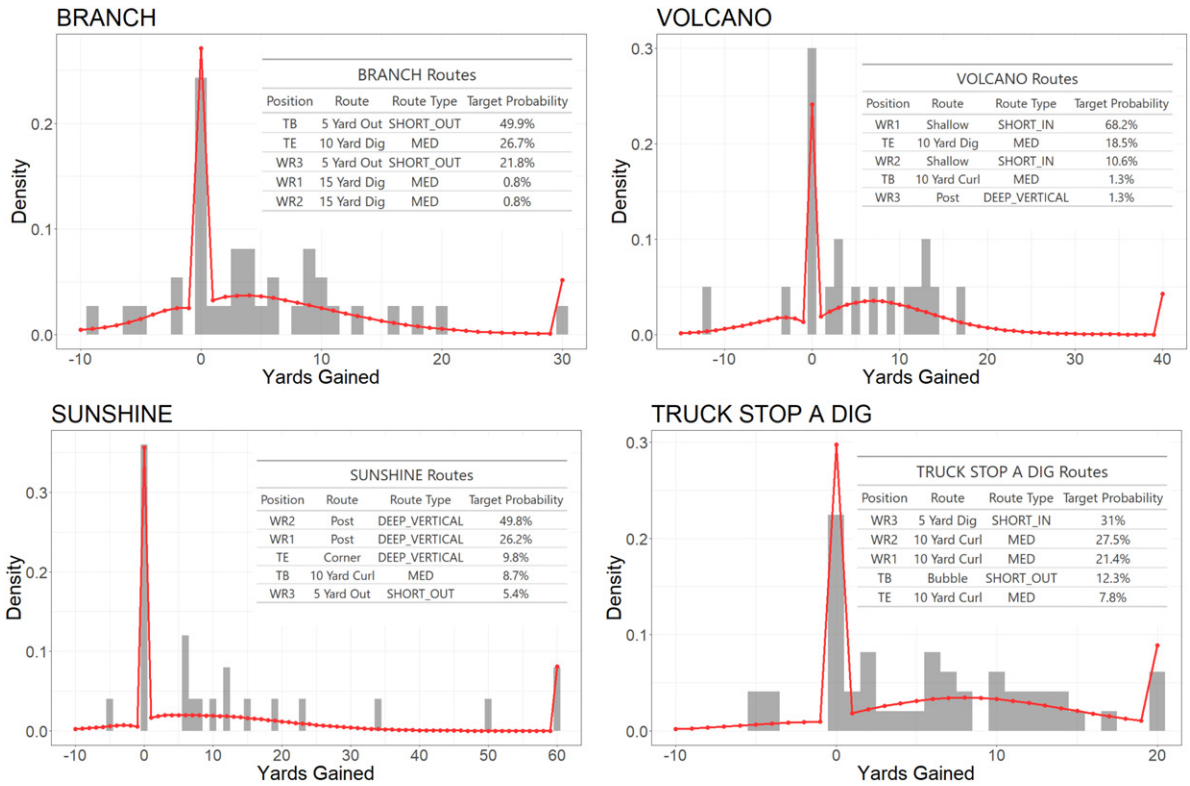

Figure 4. Average of posterior predictive densities for four different pass plays, plotted over histograms of observed outcomes for the specified plays.

The outcome for w_0 and w_t are determined, 0 yards and L_i, respectively. All the other outcomes are modeled using the generalized gamma density. We will model the $(\theta_s, \theta_1, \ldots, \theta_5)$ over all i. The i will determine the probabilities of types of outcome and the specific outcome will depend on how well the game is played. Hence, the posterior model; that is, the predictive density, for play i can be written as

$$f_i(y) = w_{0i}\, 1(y = 0) + w_{ti}\, 1(y = L_i) + w_{si} f(y \mid \theta_s) + \sum_{j=1}^{5} w_{r_j i} f(y \mid \theta_j), \tag{5}$$

with the generalized gamma posterior distributions assigned to $\theta = (\theta_s, \theta_1, \ldots, \theta_5)$ and the Dirichlet posterior for each (w_i).

4. Results and Analysis

4.1. Modeling Results

Figure 4 shows posterior predictive distributions for several pass plays. The histograms represent observed results from plays with the same base play call, and each line shows the average of several draws of the posterior predictive distributions. In these plots, we can see several nice properties of the modeling techniques. To start, we can see that the model fits are proper for each set of data, with none of the represented plays straying far from their predicted densities. Moreover, the fits are reasonable for the design of the plays, where a deep pass such as SUNSHINE has a fairly flat distribution between 0 and 60, in contrast to a shorter pass play like BRANCH having a relatively higher density focused around the 1–10 yards range and fairly low density for 20 or more yards. Furthermore, we see how a play not in the playbook such as TRUCK STOP A DIG can be adequately modeled. On this play, the receivers are told to run the TRUCK STOP concept, while the A wide receiver is tagged (given an alternative route) with a DIG route. The resulting distribution matches both the intuition of the play design and the resulting observations (pulled from plays of similar structure), showing great model robustness and flexibility.

Another nice feature of the model is the independent modeling structure and the separability of the different event outcomes, allowing us to see and examine the events contributing to the final distribution. Figure 5 shows an example of this split. Looking at the original BRANCH play, we can see that the posterior is mostly influenced by the SHORT_OUT outcome, expecting the ball to be completed to one of the two receivers running a route of this nature. In the second plot, the posterior distribution is changed by tagging the A wide receiver with a GO route, switching what was originally a SHORT_OUT route into a DEEP_VERTICAL route. This has the intended result, decreasing the relative weight of the SHORT_OUT component and adding in a DEEP_VERTICAL component and its corresponding distribution of outcomes. This is a potential way to help design and modify plays, providing insight on how results are expected to occur on an individual play and how slight modifications could potentially influence the final results.

4.2. Recommendations

Using the modeled posterior predictive distributions, we can evaluate the utility of each play using the RUM. For any given threshold of yards desired, x, we can calculate the action which

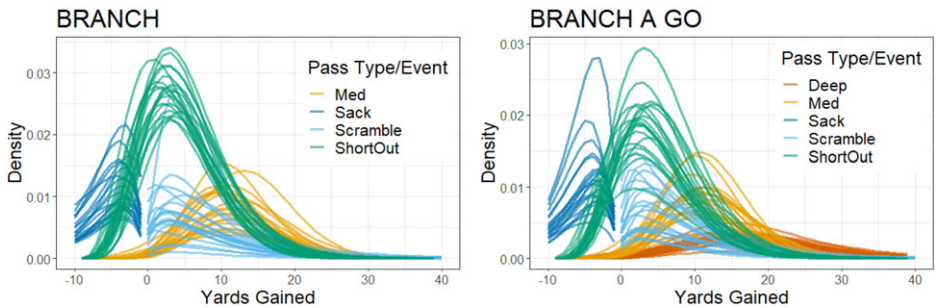

Figure 5. Draws from posterior predictive densities for two similar pass plays, split among the different continuous distributions that makeup the individual plays. Point mass pass events (interceptions, touchdowns, and incompletions) have been removed to magnify differences in areas of interest.

Play \ Threshold	1	2	3	4	5	6	7	8	9	10	12	15	20	25	30
SKEWER	59.2%	56.7%	53.9%	50.4%	46.6%	42.3%	38.4%	34.4%	30.2%	26.2%	19.9%	12.8%	7.0%	5.1%	4.5%
BACKSLASH	58.8%	56.5%	53.6%	49.7%	45.8%	41.9%	37.6%	33.2%	29.7%	25.8%	19.6%	12.4%	7.0%	5.2%	4.6%
TRUCK STOP	56.8%	54.5%	51.7%	48.7%	45.3%	42.1%	38.8%	35.3%	32.1%	28.9%	22.7%	15.5%	8.2%	5.3%	4.4%
SCORPION	52.5%	50.9%	48.9%	46.4%	44.0%	41.1%	38.2%	35.3%	31.9%	29.0%	22.8%	15.2%	8.2%	5.3%	4.4%
SUNSHINE	51.6%	49.7%	47.8%	45.7%	43.7%	41.7%	39.7%	37.7%	35.6%	33.6%	29.8%	24.5%	17.6%	12.9%	10.3%
LADDER	50.8%	49.6%	48.0%	46.2%	44.5%	42.6%	40.5%	38.6%	36.4%	34.4%	30.2%	24.7%	17.5%	13.2%	10.7%
CORN	46.7%	46.2%	45.1%	44.2%	42.8%	41.3%	39.8%	38.0%	36.4%	34.6%	31.1%	26.0%	19.3%	15.2%	13.2%
WRENCH	46.6%	45.6%	44.5%	43.6%	42.2%	41.1%	39.7%	38.0%	36.4%	34.7%	30.9%	25.9%	18.9%	14.0%	11.2%
DINNER	45.8%	44.6%	43.2%	41.8%	40.4%	38.4%	36.9%	35.2%	33.6%	31.9%	28.7%	24.1%	17.5%	13.0%	10.5%
MONOPOLY	40.5%	40.2%	39.3%	38.6%	37.8%	36.8%	35.7%	34.6%	33.4%	32.3%	29.7%	25.3%	18.8%	14.3%	11.5%

Figure 6. Optimal play calls under Random Utility Model evaluated for Presbyterian College plays for Fall 2021. Each column represents a different threshold value x, each row represents a different play or action a, and each cell represents the corresponding expected utility for an action for a given threshold for success $V_x(a)$. Here, blue and red cells indicate plays with higher and lower utilities for the given threshold, respectively.

has the largest probability of gaining the threshold amount of yards. Figure 6 shows a subset of the PC plays and their corresponding utilities under the RUM for different thresholds x.

Looking at these results, it is clear there are clusters of plays which correspond to better actions given certain desired thresholds. For example, if a coach is in a situation where he/she needs 2 yards, the model recommends the plays SKEWER, BACKSLASH, and TRUCK STOP, as they have the highest probabilities for this threshold. If he/she needs a more moderate gain, such as 6–8 yards, the recommended plays are LADDER, SUNSHINE, or WRENCH. For a larger requirement of 10 or more yards, plays CORN, DINNER, and MONOPOLY become advisable. All plays have monotonically decreasing utility for increasing thresholds for success, as it is required for the distribution function to be strictly increasing. However, the rate of decrease differs for different actions, and thus some actions become more useful under different mindsets.

Notice that some plays will see their usefulness plummet under this structure. For example, the play SCORPION is never in the top three of potential play calls under any given threshold. Assuming this subset of plays are used in similar scenarios, this result implies there is no threshold where this play would be recommended if one wished to maximize utility under a RUM framework. This does not imply the play should be completely thrown out of the playbook, but it does imply that the play has less comparative success than its peers and that it likely either needs better execution or more careful timing of when it is used to avoid wasting plays. Over time, one might conclude that the play has been called enough times to confirm that it is strictly less valuable than other plays of its type (in Game Theory terminology, the play is said to be strictly dominated by other actions (Rapoport 1966)), in which case we would recommend removing the play from the playbook altogether, unless there were some other utility structure which showed its benefits.

5. Conclusions and Discussion

The methods discussed in the article reveal unique insight and recommendations provided by the RUM. While not all decision makers think with such a mindset, from our experience (first hand) it is clear that many, if not the majority of coaches will often express binary mindsets in critical situations. For example, while a coach may not directly say "I want to gain at least x yards," he/she often will say "I want to get a 1st down," or "I want to setup a manageable 2nd down," or "I want to take a shot." All of these, while not expressed in binary language, are essentially saying the same thing: there is a certain amount of yards that will constitute success because it achieves a certain prescribed goal in the mind.

This does not discredit the merit of long-term decision models. The idea of maximizing expected utility has been shown to have positive behavior throughout the literature. However, decision makers often do not behave according to long-term optimal policies, and therefore the RUM fills a gap in the literature. It is not difficult to explain to a rational decision maker the benefits of a long-term utility structure. However, on a critical fourth and goal situation where the offense has opted not to kick a field goal, coming short of scoring a touchdown would be viewed as a failure by the coach and the players. One could argue that leaving the opponent with the ball on their goal line versus another position where the resulting offense is closer to scoring would be more advantageous, but we argue that this is often not how those on the field view the result of their actions, but rather is the carefully examined work of the isolated analyst who views the game in terms of expected points. Indeed, it is

more likely that the coach chooses a play in a way that he/she believes maximizes scoring probability, and therefore we wish to represent that mindset.

There are some areas for extensions. However, all improvements need to be balanced with the complications they induce. While including finer details could potentially increase modeling accuracy, it would reduce the ability for a coach to make decisions on his/her feet, and therefore additional information may not prove helpful. Some potential improvements would be the inclusion of penalties, a more robust representation of turnover behavior, and the incorporation of run plays. There are many opportunities for the analyses of these methods, including a broader analysis of pass concepts, personnel groupings, or even defensive alignment and play calling. Finally, this work currently ignores the adversarial and sequential nature of the sport, assuming the parameters of the probability distributions are constant throughout the game and against all opponents. One could drop this set of assumptions, allowing for dynamic determination of posterior distributions based on previously observed results and the quality of opponents.

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ORCID

Preston Biro https://orcid.org/0000-0003-3405-6881

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